Shosuke Kiami

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Education

University of California, Berkeley, Berkeley CA — Aug 2025 - Present
Ph.D. Applied Mathematics

University of Washington, Seattle WA — Sep 2020 - Jun 2024
B.S. in Computer Science, B.A. in Mathematics
Cumulative GPA: 3.98/4.00
Honors: Magna Cum Laude, Phi Beta Kappa, Annual Dean's List

Publications

Shosuke Kiami. Lossless Convexification for Linear Systems with Piecewise Linear Controls. arXiv:2411.18004.
- Under review at *American Control Conference 2026*.

Alex Albors, François Clément, **Shosuke Kiami**, Braeden Sodt, Ding Yifan, Tony Zeng. Approximately Jumping Towards the Origin. arXiv:2412.04284.
- Under revision at *Involve Undergraduate Mathematics Journal 2025*.

Chris Jennings-Shaffer, David H Rich, Matthew Macaulay, Michael D Karcher, Tanvi Ganapathy, **Shosuke Kiami**, Anna Kooperberg, Cheng Zhang, Marc A Suchard, Frederick A Matsen IV. Finding high posterior density phylogenies by systematically extending a directed acyclic graph. arXiv:2411.09074.
- Accepted to *Algorithms for Molecular Biology 2024*.

Zoey Chen, **Shosuke Kiami**, Abhishek Gupta, Vikash Kumar. GenAug: Retargeting behaviors to unseen situations via Generative Augmentation. arXiv:2302.06671.
- Accepted to *Robotics: Science and Systems 2023*, awarded Best System Paper Finalist.

Michael Duan, **Shosuke Kiami**, Logan Milandin, Johnson Kuang, Michael Saugstad, Maryam Hosseini, and Jon E. Froehlich. Scaling Crowd+AI Sidewalk Accessibility Assessments: Initial Experiments Examining Label Quality and Cross-city Training on Performance. dl.acm.org.
- Accepted to *ACM SIGACCESS Conference on Computers and Accessibility 2022*.

Research Experience

Independent Research, UC Berkeley, Berkeley CA — Sep 2025 - Present
Advisor: Jelani Nelson
- Exploring various applications of stochastic processes to streaming algorithms.
- Developing a new $O(\log n)$ bit super heavy hitter algorithm with success probability $1-\varepsilon^k$, making progress towards solving ℓ_2 heavy hitter in optimal $O(\log(1/\varepsilon^2)\log(n))$ space.

Independent Research, UC Berkeley, Berkeley CA — Sep 2025 - Present
Advisor: Satish Rao
- Independent reading on various aspects of the KLS conjecture, including Bo'az Klartag's new bound on the isoperimetric constant for high dimensional convex bodies.
- Re-derived the proof for the stopping time of Eldan's stochastic localization process.

Independent Research, UC Berkeley, Berkeley CA — Jun 2025 - Aug 2025
Advisor: Venkatesan Guruswami
- Developed new exponential-time approximation algorithms for various constraint satisfaction problems such as Max-kLin.

Autonomous Controls Lab, UW, Seattle WA — Jan 2024 - Jun 2025
Advisor: Behçet Açıkmeşe
- Single-authored a paper (arXiv) introducing a new optimal control algorithm that extends lossless convexification theory from stepwise to piecewise-linear parameterized controls.
- Proved guarantees of the above algorithm using results from books such as *Convex Analysis* by Borwein & Lewis and *Numerical Optimization* by Nocedal & Wright.

Washington Experimental Mathematics Lab, UW, Seattle WA — Sep 2024 - Dec 2024
Advisor: Stefan Steinerberger
- Co-authored a paper (arXiv) examining a class of dynamical systems in the complex plane.
- Discovered and proved results for the stochastic variant, including our main theorem regarding the existence and expectation of the stationary measure in higher dimensions.
- Presented at the *Northwest Undergrad Math Symposium 2024*, receiving 2nd place prize.

WEIRD Robot Learning Lab, UW, Seattle WA — Jan 2023 - Jun 2023
Advisor: Abhishek Gupta
- Collaborated on a paper (arXiv) introducing generative models as an effective data augmentation technique for robot learning, winning Best System Paper Finalist at *RSS 2023*.

Matsen Group, Fred Hutchinson Cancer Center, Seattle WA — Jun 2021 - Sep 2021
Advisor: Frederick Matsen
- Contributed to a paper (arXiv) introducing the sDAG: a data structure that compactly represents a collection of phylogenetic trees in order to perform Bayesian inference.
- Developed fast tree rearrangement algorithms necessary for MCMC steps.

Makeability Lab, UW, Seattle WA — Jun 2020 - Jun 2022
Advisor: Jon Froehlich
- Started an initiative to use deep learning for automatic sidewalk accessibility evaluation.
- Co-authored a paper (dl.acm.org) examining model performance across international cities.

Work Experience

Software Engineering Intern, Apple, San Diego CA — Jun 2023 - Sep 2023
- Introduced an improved adaptive bitrate algorithm reducing runtime from $O(n^2)$ to $O(n)$.

Software Engineering Intern, Microsoft, Redmond WA — Jun 2022 - Sep 2022
- Designed a custom image segmentation algorithm reducing runtime from 80 ms to 0.5 ms.

Teaching Experience

MATH 54 - Linear Algebra, TA, Berkeley CA — Sep 2025 - Dec 2025
- Led 6 hours of weekly discussion sections on linear algebra and differential equations.
- Wrote/proctored exams, graded assignments, and held weekly office hours.

Fusion Math, Math Tutor, Seattle WA — Jan 2025 - Jun 2025
- Taught lessons covering curriculum in AP Calculus.

CS Tutor, Self-employed, Seattle WA — Mar 2022 - Jun 2022
- Taught lessons covering curriculum in AP Computer Science.

Seattle Music Partners, Cello Teacher — Sep 2016 - Jun 2020
- Taught cello lessons to students at underserved elementary schools.

Personal Projects

Euclideate — Nov 2024 - Present
- A highly-optimized C++ algorithm that solves straightedge + compass construction problems, with the goal of finding/proving the minimal construction of the 17-gon (GitHub).

DanceTime — May 2021 - Jul 2023
- A C++ *Just Dance* remake with a custom gesture scoring algorithm using pose estimation, conditional filtering, polynomial regression, and numerical optimization (GitHub).

AlphaFour — Jun 2022 - Jul 2023
- A Python *Connect 4* AI that learns via self-play deep reinforcement learning (GitHub).

GeoKnowr — Nov 2022 - Dec 2022
- A Python *GeoGuessr* AI that reliably guesses within 2000 km of the ground truth (GitHub).

Extra-curriculars

UW Formula Motorsports, Engineer — Jan 2023 - Jun 2024
- Designed the team's first driverless steering system (motor + custom rack and pinion).